

DESIGN OF THE PROTOTYPE FAST REACTOR

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1. Introduction

This paper describes the main design features of the AEA's 250 MW(e) sodium cooled prototype fast reactor (PFR) as it stood at the end of 1965. Other aspects of the design are described in refs. (1) and (2). Although the many alternatives considered in the development of this design are not discussed in the paper, the philosophy leading to the chosen arrangement is outlined. The design is supported by development work carried out in industry and in AEA laboratories.⁽³⁾ A diagrammatic arrangement of the complete system is shown in Fig. 1 and a list of the main parameters appears in the Appendix.

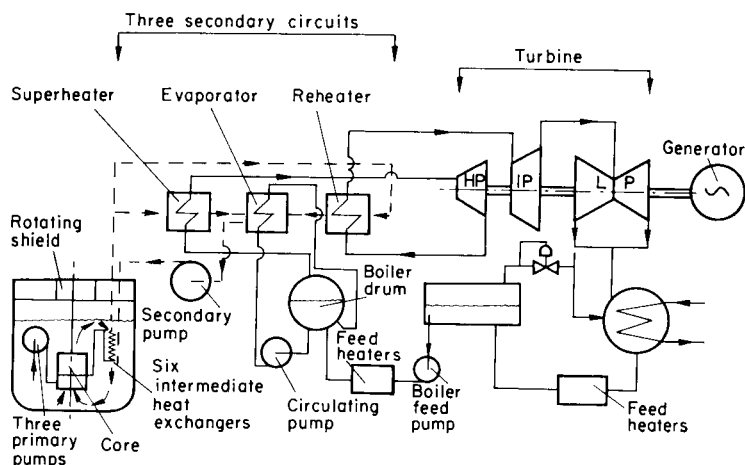


FIG. 1. Diagrammatic arrangement of plant.

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The power output of the PFR is based on the requirement that it should be the smallest reactor capable of providing, with confidence, the information necessary for the installation and operation of commercial fast reactors in the United Kingdom in the late 1970's or early 1980's. After consideration of the fuel development, physics and safety aspects, and finally of the engineering features of the reactor, it was concluded that a minimum power output of 600 MW(th) was necessary. The capital cost of the station has been minimized, although the inclusion of experimental facilities has incurred some additional expenditure.

The main features of the design are the use of ceramic fuel elements, the layout and containment of the single tank primary circuit, the fuel handling arrangements, the use of single wall steam generators and the adoption of temperatures which permit modern steam conditions of 2300 lb/in² g at 960–1050°F (162 kg/cm² at 516–566°C), to be used.

2. Design Philosophy

(a) Temperatures

The coolant outlet temperature is limited, with the first charge, by the maximum fuel clad temperature of 650°C (1200°F), and in the reference design for a mixed oxide core is in the range 560–600°C (1040–1192°F). The temperature at inlet to the core, 400–430°C (752–806°F) was determined from parametric surveys based on a 1000 MW(e) reactor.

(b) Core Design

Upward flow of coolant through the core was chosen because of the importance attached to avoiding gas entrainment at the free surface above the core and to guaranteeing reasonable distribution of flow over the full range of reactor output and to ensure natural circulation. It was also considered important to have a low pressure blanket gas over the core.

The core is divided into subassemblies for the purposes of handling and support of the fuel pins. The length and cross-sectional area of the subassembly was selected after a detailed study of the fuel cycle economics of 1000 MW(e) reactor systems, and the PFR subassembly is designed so that, with the enrichment suitably adjusted, it can be used in a commercial reactor without modification. There are two enrichment zones in order to achieve a reasonable radial form factor; an arrangement which appears suitable for a commercial reactor. The support of the subassemblies is such that they do not rattle in their spigots and they are arranged to distort into a less reactive configuration under the influence of radial temperature gradients.

Although the reference core design is based on mixed oxide fuel the reactor will accept carbide and probably other types of fuel either in small experimental quantities or as complete cores. Vented fuel assemblies can be accepted and the blanket cleanup system is capable of dealing with a complete core of vented fuel.

Control is by movement of absorber since fuel control rods would be too large. The absorber is divided between control and shut-off rods and since the only means of direct access to the core is via the rotating shield, which is moved during fuel handling operations, all absorber rods are detached from their actuators and left in the core during fuel handling.

(c) Primary Circuit

Considerable effort has been spent in comparing different arrangements of primary circuit. This has resulted in the choice of a single tank for containing all the primary sodium since this gives a more compact arrangement and avoids the difficult problems inherent in the design of pipework in a system of separate tanks. The single tank arrangement appears to have a lower capital cost and is easier to contain, within a strong vault and roof, against pressure which might result from a core incident. A single tank, which can be designed without penetrations or attachments, achieves the highest possible integrity against leakage of coolant. Against these advantages must be set the difficulties associated with shielding the pumps and heat exchangers and the problems of site construction.

It was considered that the prototype should give experience of operating multiple loops in parallel. Kinetic analysis showed that the failure of one pump in three gave an acceptable transient, so three primary loops were adopted. The size of a primary pump (approximately 18,000 gal/min, 82,000 l/min) represented a reasonable intermediate step between the experimental pump already under test and the larger pumps envisaged for later reactors. A comparison of mechanical and electromagnetic sodium pumps indicated that the former had greater potential for extrapolation to large outputs and the mechanical pump had a clear advantage in efficiency. Mechanical pumps were, therefore, chosen for the prototype reactor.

All primary circuit components are supported from the vault roof and can be withdrawn for maintenance. Sufficient internal shielding is provided to allow components, such as pumps and heat exchangers, to be worked on once adhering primary sodium has been removed.

(d) Safety Principles

The PFR safety principles, discussed in more detail in section 4, can be summarized as follows:

- (i) Loss of coolant from the core is discounted on the basis of rigorous

design of the primary tank with all penetrations placed in the roof, and provision of a completely separate leak jacket; the arrangement of the primary circuit and leak jacket being such that they are contained in a concrete vault below ground level.

- (ii) Temperature transients, due to coolant flow failure, are controlled by tripping the reactor (guaranteed by a multiplicity of trip parameters) and provision of sufficient pump inertia to restrict the maximum temperature to acceptable values. The large volume of sodium in the primary tank being directly cooled at all times by a number of independent natural convection loops.
- (iii) Reactivity transients are controlled by limiting the maximum rate of reactivity insertion, assuming all absorber is moved simultaneously to a value such that the transient can be terminated by reactivity trips without exceeding fuel failure temperatures.
- (iv) Propagation of certain subassembly faults cannot yet be ruled out and therefore the inherent strength of the primary tank, leak jacket and vault has been exploited to provide a containment system. A series of model tests has demonstrated the adequacy of this containment arrangement.

(e) Secondary Circuits

A secondary sodium system incorporating three circuits in parallel is shown diagrammatically in Fig. 1 and is introduced between primary sodium and the steam plant to ensure that the core cannot be blocked by sodium–water reaction products in the event of a steam generator tube failure and that active primary sodium is not involved. The three secondary loops are not interconnected in any way so that leakage can result in the draining of one loop only.

The steam generator units are simple shell and tube heat exchangers with provision for isolating the effects of a sodium–water reaction and dumping the reacting fluids.

(f) Fuel Handling

One of the most difficult decisions in the conceptual design of PFR lay in the choice of the refuelling system. Since the choice of refuelling system can have a marked effect on the reactor layout, availability, core instrumentation, and experimental facilities, requirements of a commercial reactor system were studied very carefully before settling the PFR arrangement. This study indicated that there was a small economic incentive in favour of an on-load refuelling system for a commercial fast reactor. However, the complexity of the mechanisms and problems of core design involved in such an arrangement were such that it was considered that the loss in reliability of the system

would almost certainly reduce the calculated gain. Additionally, there were the problems of controlling reactivity and coolant flow during the refuelling operation. For these reasons on-load refuelling was rejected for PFR.

Because of the problems in transferring highly rated subassemblies from a sodium system shortly after shutdown, it was considered essential that subassembly transfer should take place under sodium until the decay heat had fallen to such a level that reduced cooling could be accepted. This requirement had a considerable effect on the depth of the primary tank.

3. Description of Design Features

(a) Core

The reference core is made of hexagonal subassemblies (see Fig. 2), 5.6 in. (14.2 cm) across the flats, 12 ft 7 in. (381 cm) long containing 325 UO_2 - PuO_2 fuel pins. Each fuel pin is 0.23 in. (0.585 cm) o.d. with a 0.015 in. (0.038 cm) thick wall and is generally as described in ref. (1). The pins in the subassembly are supported by grids on a 4 in. (10.15 cm) pitch, over the 36 in. (91.5 cm) core length and at either end of the 9 in. (22.9 cm) long axial breeder. Below the lower breeder section there is a 47 in. (119 cm) long fission gas reservoir slightly smaller in diameter than the in-core section of the pin in order to reduce the coolant pressure drop. The ratio of core to axial blanket length is under review at the time of writing and may be changed. An alternative design of pin is being considered in which the fission product gas is released into the coolant without pressurization of the pin.

The subassembly is fixed in a grid by a three point support. The lower bush in the grid is offset away from the control rod position so that the spike of the element is compelled to bend in this direction and force the uppermost support against the wear faces provided on the control rod guide tubes. Hydraulic forces are balanced by a conventional hydraulic hold-down. No mechanical hold-down is provided though the instrument tubes suspended from the top shield will limit the extent to which an element may rise.

The subassemblies are arranged in a lattice as shown in Fig. 3. The inner zone of low enrichment contains thirty elements and the outer some forty-eight. The core is surrounded by a radial core reflector to improve the radial form factor, reduce the fissile inventory, and increase the internal breeding ratio. This reflector is followed by a single row of radial breeder subassemblies to represent the two or three rows that would probably be employed in a commercial fast reactor. Outside this again is a breeder reflector to reflect escaping neutrons and improve the breeding. The radial reflector and blanket design is not yet finalized and alternative arrangements, currently being considered, will be examined after the PFR mockup being tested in ZEBRA has been assessed.

Control is obtained by five tantalum control rods, which can be moved in any combination from one to five, so as to even out variations in the rate of reactivity addition. Although tantalum is currently proposed, enriched boron

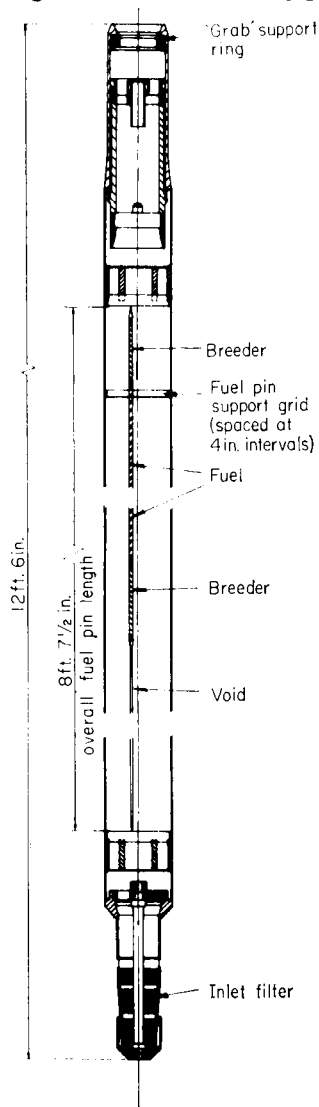


FIG. 2. Fuel subassembly.

carbide may be used if satisfactory long term radiation results can be obtained. Six shut-off rods of enriched boron carbide at the core-reflector boundary are used to maintain sufficient sub-criticality during fuel handling and as a back-up emergency shutdown system.

The control rod consists of a shell and tube assembly with sodium flowing through the tubes cooling the tantalum plates. The shut-off rod consists of an assembly of stainless steel cans containing enriched boron carbide within a stainless steel hexagonal wrapper plate. No forced cooling is supplied to

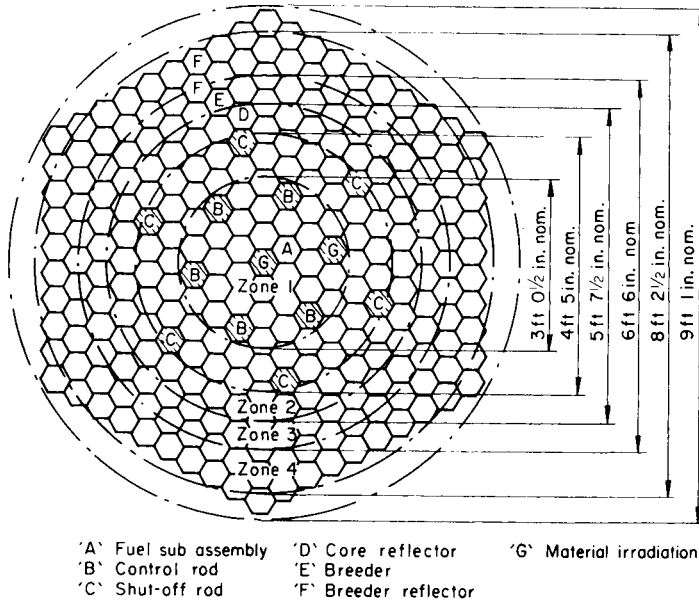


FIG. 3. Core plan.

TABLE 1

	Control rods	Shut-off rods
Number off	5	6
Type	Tantalum plate	Boron carbide cluster
Weight	300 lb (136 kg)	140 lb (63.5 kg)
Stroke	42 in. (107 cm)	54 in. (137 cm)
Speed, rod 18 in. withdrawn	1.636 in./min (4.15 cm/min)	3.75 in./min (9.5 cm/min)

the shut-off rods. Details of the control and shut-off rods are summarized in Table 1 above.

It is anticipated that the only maintenance required will be to the operating mechanism, i.e. the section above and including the magnet, and the assembly has been designed so that this section may be readily removed in a flask.

Replacement of absorber subassemblies within the core is carried out in the same manner as the refuelling process and provision is made for the spent rods to decay in the storage drum within the reactor vessel or in the area surrounding the radial breeder. The control and shut-off guide tubes suspended from the rotating shield also house thermocouple conduits each of which serves a group of six subassemblies surrounding the rod.

(b) Primary Circuit

The whole of the primary circuit is contained within the primary tank which is a cylindrical vessel 40 ft (1220 cm) diameter by 50 ft (1520 cm) deep fabricated from $\frac{1}{2}$ in. (1.27 cm) stainless steel. It is suspended, as shown in Fig. 4, at the outer periphery from the roof support structure. This type of construction enables the main components of the circuit to be suspended symmetrically from the roof, thus keeping the stresses on the sodium containment system to a minimum. A separate low alloy steel leak jacket surrounds the reactor tank and prevents excessive fall in liquid metal level should the primary circuit fracture. The leak jacket is contained within a reinforced concrete vault situated below ground level.

When the reactor is operating, sodium at 400°C (752°F) is drawn from the bulk sodium in the tank by three mechanical pumps in parallel and delivered at 9.2 kg/cm² abs. to pipes which pass to the core grid. This grid, together with the reactor jacket and thermal shield rods, is supported as shown in Fig. 4 by a structure attached to the roof. After passing through the core the liquid metal at 560–600°C (1040–1112°F) flows through the neutron shield rods to the intermediate heat exchanger. Coolant emerging from the intermediate heat exchangers at 400–430°C (752–806°F) is distributed around the heat exchanger shells and sweeps the tank surface before flowing to the pump suction. In this way the risk of stratification of hot sodium is reduced. The argon gas blanket covering the primary sodium system is pressurized to about 1 lb/in² (0.07 kg/cm²). The pressure drop through the primary heat exchanger is provided by a difference in free surface level between the coolant at the heat exchanger inlet and the main tank.

The $\frac{1}{2}$ in. (1.27 cm) thick stainless steel reactor vessel is designed to the appropriate AEA standards for radioactive sodium, all welds being radiographed. High integrity is obtained through simplicity of design, there being no penetrations or nozzles in the vessel below sodium level. The main design problem is to limit the thermal stress at the top strake where the vessel is hung from the roof. This problem is overcome by the use of graded insulation and external cooling to give an acceptable axial temperature gradient in the vessel wall between the free surface and the roof. A low voltage heating system, together with strain gauges, thermocouples, and sodium leak detection equipment, is fitted to the outside of the reactor vessel. This equipment

is incorporated into a mild steel frame which serves to support the lagging and is suspended from a series of pads welded to the top section of the vessel.

A 20 ft (610 cm) diameter free standing cylindrical steel skirt sits on the core support structure and has three distribution compartments radiating from the top end into which the heat exchangers are fitted. The internal

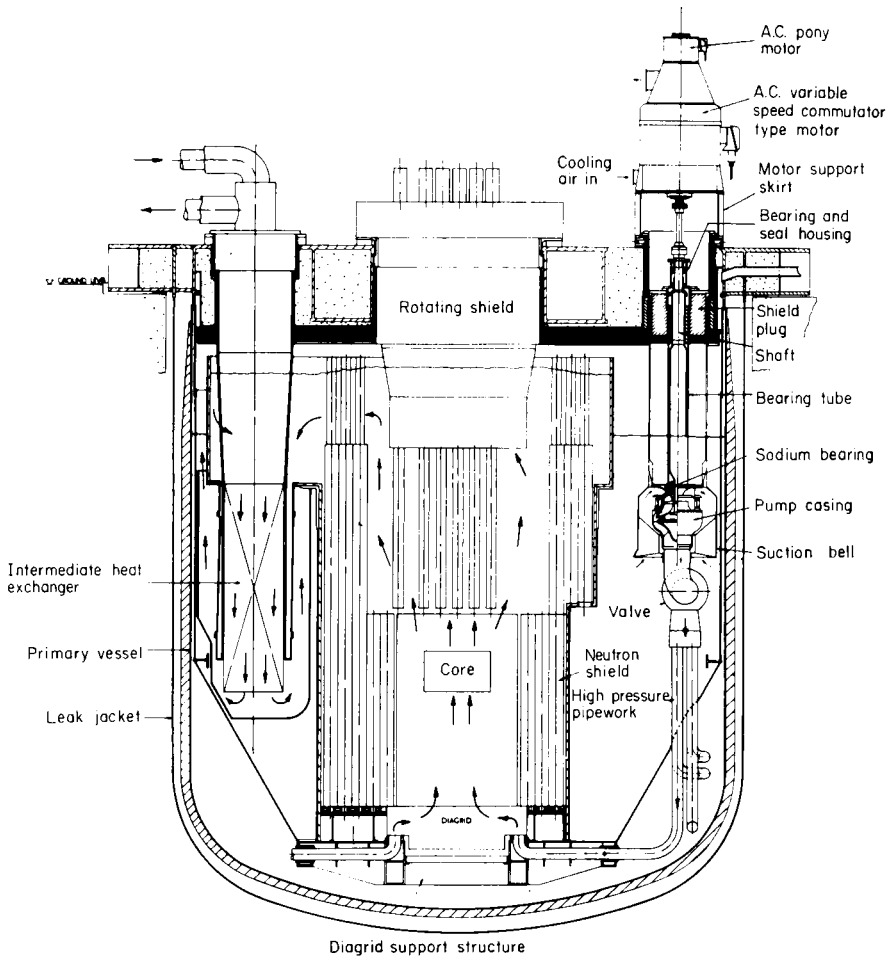


FIG. 4. Primary vessel sectional elevation through a pump.

surface of this reactor jacket is clad with stainless steel insulation and the structure as a whole distributes the hot sodium through the neutron shield rods to the intermediate heat exchangers.

The thermal insulation is required to protect the reactor jacket structure from thermal shock and steady state contact with the high temperature sodium and also to limit the amount of heat transfer between the hot outlet

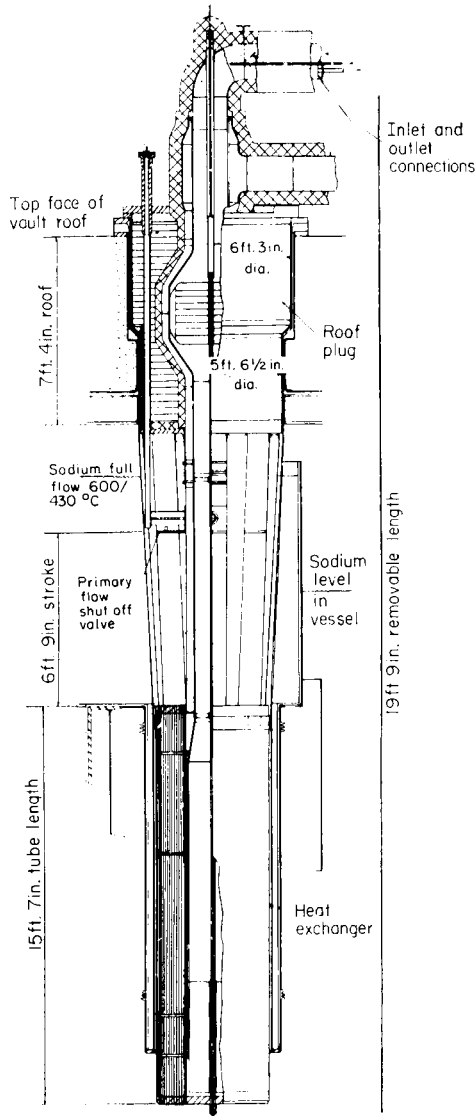


FIG. 5. Intermediate heat exchanger.

sodium and the bulk sodium. Layered insulation has been specially developed for this purpose and each layer consists of two thin sheets of stainless steel welded together to form a quilted pocket construction with trapped gas filling the pockets. These panels are then fabricated into independent layers which fit closely to the reactor jacket.

The grid and the surrounding neutron shield is positioned and supported

from the grid support plate which is suspended from a cylinder attached to the roof as shown in Fig. 4. This cylinder is 6 in. (15.2 cm) smaller in diameter than the reactor vessel and changes to a conical shape at a ring beam 24 ft (730 cm) below the roof. The cylinder is castellated where it is connected to the roof to take out the hoop component of the thermal stress.

The neutron shield is supported from the top of the grid support plate by means of cylindrical steel supports carrying location plates. The support system is designed so that deflexions caused by the shield rods are not transmitted to the grid and inlet pipework joints. This is achieved by fitting the grid on stellited rocking pads and making the pipework independent of the support plate.

The six intermediate heat exchangers shown in Fig. 5 are of the counter-flow shell and tube type with 1540 15 ft (457 cm) long, $\frac{1}{2}$ in. (1.9 cm) o.d. tubes enclosed in a 4 ft (122 cm) diameter shell. Flexibility between individual tubes and the shell is provided by sinusoidal bends in the tubes. Primary coolant flows through the tubes and secondary sodium flows on the shell side. There are no baffles in the shell and in order to locate the tubes and minimize vibration a number of grid supports are positioned along the tube length. The heat exchanger, which is removable, is supported from a flanged plug in the reactor vessel roof. A programme of development work covering vibration, fretting, tube to tube plate welding, and gas entrainment in the intermediate heat exchanger has been put in hand.

The three primary pumps are of a mechanical, vertical shaft mixed flow type with a free gas surface. The pumps are driven by fixed speed motors through variable speed hydraulic couplings and employ an oil lubricated face seal to provide a leak tight barrier between the cover gas and the atmosphere. An alternative variable speed drive involving an a.c. commutator motor is being considered. The pumps, which are interchangeable, are supported from a flanged plug in the roof. The diameter of the pump, which is an important factor in determining the overall diameter of the primary vessel, is minimized by adopting a mixed flow impeller in a bowl type casing. The shaft is supported by a taper-roller bearing at the top of the plug and a sodium lubricated bearing just above the impeller. The pump is of simple rugged construction with as few parts as possible and careful attention has been paid to the problems of removal and maintenance. The reference design was preceded by the design and development by the AEA of a prototype pump which has been in operation in the engineering laboratories under simulated reactor conditions.

(c) Secondary Circuit and Steam Generator

There are three separate secondary loops, each loop being connected to two intermediate heat exchangers. In the reference design the hot sodium

(530°C) (985°F) passes via an isolation valve to the high temperature end of the steam generator where the flow is divided between the reheater and superheater. After passing through these units in parallel the flow is recombined and enters the evaporator at 455°C (851°F). After leaving the evaporator at 370°C (698°F) the sodium returns to the intermediate heat exchanger via an expansion tank, isolation valve, and secondary pump.

The entire secondary loop outside the steam generator is fabricated in type 321 stainless steel with full radiographed butt welding throughout. All pipework and vessels containing sodium are trace heated.

The secondary circuit pressure drop is approximately 45 lb/in². (3.2 kg/cm²). Flow at full design load is approximately 15,000 gal/min (68,000 l/min) in each loop. The pumping power of approximately 700 b.h.p. (522 kW) is provided by a 925 rev/min mechanical vertical spindle pump similar in arrangement to the primary pump. A minimum speed range from 100% down to 20% is specified and will be provided either by an induction motor driving through a hydraulic coupling or by an a.c. commutator motor.

There are free sodium surfaces at the inlet and outlet ends of the evaporator, superheater, and reheater units and also in the expansion tank and above the isolation valves and secondary pumps. It is currently anticipated that these free surfaces will be blanketed with argon gas and sealed during initial heating so that the sodium levels adjust themselves according to the static pressure of the sodium. There is an argon supply and vent line to each of the gas volumes.

The steam generators have been designed thermally in accordance with the parameters in the Appendix. To allow for a possible increase in the core outlet temperature to 600°C (1112°F), however, materials have been chosen and the equipment stressed to operate with a maximum secondary sodium temperature of 580°C (1076°F) and a sodium temperature of 470°C (878°F) at the inlet to the evaporator. These temperatures would permit steam temperatures of 566°C (1050°F) at outlet from both superheater and reheater.

The three steam generators, each of which consist of a superheater, reheater, and evaporator unit, are located in separate bays in the steam generator building. All units are of similar arrangement but materials and dimensions differ. A superheater is shown in Fig. 6. Hot sodium enters at the bottom and passes upwards along a central duct. The flow is then turned through 180° so that it passes down an inner annulus housing one limb of the steam-water "U" tubes. At the bottom the flow is turned through 180° and passes upwards through the outer annulus which houses the other limb of the "U" tubes. The sodium leaves the shell through ports into a collection annulus. The water-steam enters the tubes via an annular header and flows through the "U" tubes to the annular outlet header, the steam flow being in counter-flow to the sodium in the high temperature units and in parallel flow with the sodium in the evaporators.

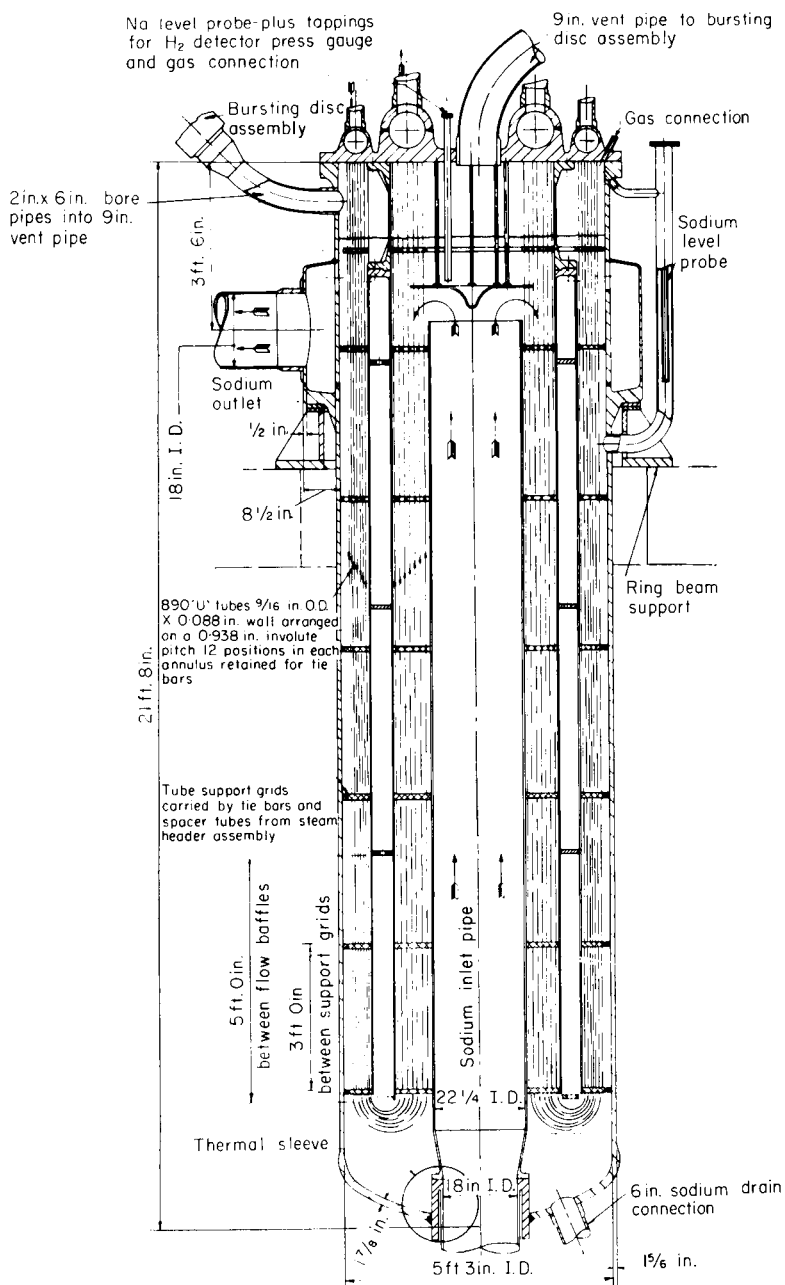


FIG. 6. Superheater.

Each steam generator has its own steam drum which receives feed water direct from the feed heating train, there being no economizer. Water is drawn from the bottom of the drum, forced through the evaporator and the mixture of steam and water returned to the drum for separation. The dry saturated steam is tapped from the top of the drum and passes to the superheater. The recirculation ratio through the evaporator is 4:1.

The high temperature units are fabricated in stainless steel throughout. Carry-over from the drum may result in a slightly wet condition near the superheater inlets. Reliance is placed on a full flow water treatment to eliminate the possibility of chloride stress corrosion in this region.

The evaporator shell is fabricated from 2½Cr1Mo ferritic steel and the tubes also will be of a ferritic steel. Tests have indicated that the removal of carbon from ordinary 2½Cr1Mo steel may be excessive and, for the tubes, consideration is being given to this steel stabilized with titanium-niobium or alternatively the 9Cr1Mo ferritic steel which does not decarburize significantly.

To relieve pressure and remove waste products created by a sodium-water reaction, following a tube failure, bursting disks are provided through which the products are vented to atmosphere. The integrity of the tubes is monitored in each unit by hydrogen detectors. In the event of failure occurring, provision is made for automatically dumping the water and sodium.

The design of steam generators has been analysed to indicate problem area and development work has been initiated in tube to tube plate welding, tube-support fretting, vibration, gas entrainment, and finally sodium-water reaction.

(d) Sodium Purity Control and Storage

The entire contents of the primary and secondary circuits can be accommodated in external mild steel tanks. This permits the accumulation of sodium during the construction phase at an economic rate. The tanks are lagged and trace heated and shielding can be placed round them if it is required to dump active sodium after operation has commenced. The delivered sodium will be filtered on its way into the tanks where further settlement of some of the impurities is expected to take place. Connexions are provided for cold trapping if this is required.

The basic methods of impurity control for both primary and secondary sodium are the cold traps which precipitate oxides and hydrides by reducing the sodium temperature below its saturation value for these constituents. It is believed that other impurities are precipitated out with the sodium oxide. Sodium is extracted from the primary tank and passes through a regenerative heat exchanger which reduces its temperature to approximately 140°C (284°F). The sodium enters the top of the externally cooled cold trap

vessel and flows downwards through a 6 in. (15.25 cm) cooling annulus where it reaches a minimum temperature of 110°C (230°F) before passing upwards through the central section which contains the collecting basket of stainless steel wire mesh. The precipitated impurities are collected on the mesh. The temperature at inlet to the basket is expected to be some 25°C (77°F) above the saturation temperature for sodium oxide and the impurity level of the sodium at outlet from the trap will be reduced to 5–10 ppm. The primary cold trap loop contains a 100 gal/min (455 l/min) pump delivering 66 ft (2013 cm) head and shielded dump tank so that active sodium from the cold trap loop can be drained. Associated with the primary cold trap are a number of monitoring facilities which will probably include:

- (i) a sampling facility for removing crucibles which contain representative samples of primary coolant for laboratory analysis;
- (ii) an experimental corrosion facility for simulating reactor sodium conditions of velocity and temperature into which steel samples can be inserted and removed;
- (iii) a plugging meter which monitors the temperature at which precipitated impurities begin to block an orifice.

The primary cold trap loop is situated in a shielded vault, access being permitted only when the sodium has been drained to the dump tank and sufficient time has elapsed for residual activity to decay. Considerable attention has been paid to the problems of maintenance of this facility since it is anticipated that fission products from fuel failures or from vented fuel arrangements will tend to accumulate in this loop. The secondary cold trap loop is also of 100 gal/min (455 l/min) capacity but is unshielded. Both loops are constructed throughout of stainless steel and are fully trace heated.

(e) Steam Turbine Plant

The turbo-generator is a conventional multi-cylinder tandem machine with single reheater, comprising H.P., I.P., and two double flow L.P. cylinders, the rotors being solidly coupled. The set is designed to operate at a stop valve temperature of 566°C (1050°F) and pressure of 2300 lb/in² (162 kg/cm²). With a nominal heated input of 600 MW the output at the generator terminals would be approximately 270 MW. It is anticipated that the set may initially have to operate at lower temperature and with a stop valve temperature of 513°C (955°F) the anticipated output is approximately 260 MW. The set is capable of operation over this range depending upon the achievable fuel temperatures.

There are eight feed heating stages, three are H.P. surface heaters and five direct contact. The condenser is arranged longitudinally under the L.P.

cylinder. Double tube plates are used to reduce contamination of the condensate.

A dump system capable of condensing approximately 10% full load steam to the turbine is provided for flexibility during start-up and load changing. The main condenser is used for this purpose and the temperature and pressure of the dump steam is reduced by a series of pressure reduction valves and desuperheaters. There is a single 100% capacity feed pump driven by a steam turbine which is normally fed by bled steam but can accept live steam during start-up or periods of low load on the main turbine. There is a 10% steam driven emergency feed pump. A full flow condensate cleanup plant is provided consisting of resin-bearing filters with means of introducing additional capacity to prevent carry-over of chloride in the event of a condenser tube failure.

The generator is a two pole, 3000 rev/min, hydrogen-water cooled machine generating at approximately 17,000 V, three phase, with a current rating of approximately 10,000 A.

(f) Fuel Handling

The fuel handling system is indicated in Fig. 7. It involves transfer of spent fuel from the core and its replacement with new fuel from the store within the reactor vessel, and is carried out with the reactor off load. The removal of the decayed fuel and its replacement with new fuel from the external store can be carried out with the reactor on power. Considering a similar system applied to a 1000 MW(e) commercial reactor and using information from fuel handling in the Dounreay fast reactor, it is estimated that the necessary fuel interchange can be carried out with a weekend shutdown every 3 months. The refuelling system employs three major units, a single rotating shield mounted concentric with the core, a charge machine having radial movement across the rotating shield, and a fuel transfer rotor located within the primary reactor vessel adjacent to the core. A combination of the movements of the rotating shield and charge machine enable any subassembly position in the core, radial breeder, and reflector to be selected. The charge machine transfers spent fuel from the core to the storage drum, all movement of fuel within the reactor vessel being carried out beneath the sodium level. The fuel cycle currently proposed for the reactor requires the replacement of thirteen fuel subassemblies about every 80 days. This can be accomplished within a 36-hour shutdown period.

The need to rotate the shield during refuelling necessitates the disconnection of control and shut-off rod actuating gear from the absorbers, the latter remaining in the core during refuelling operations. Final checking that the disconnections have been made is carried out by sweep arms incorporated in the rotating shield. The rotating shield, 12 ft (366 cm) in diameter,

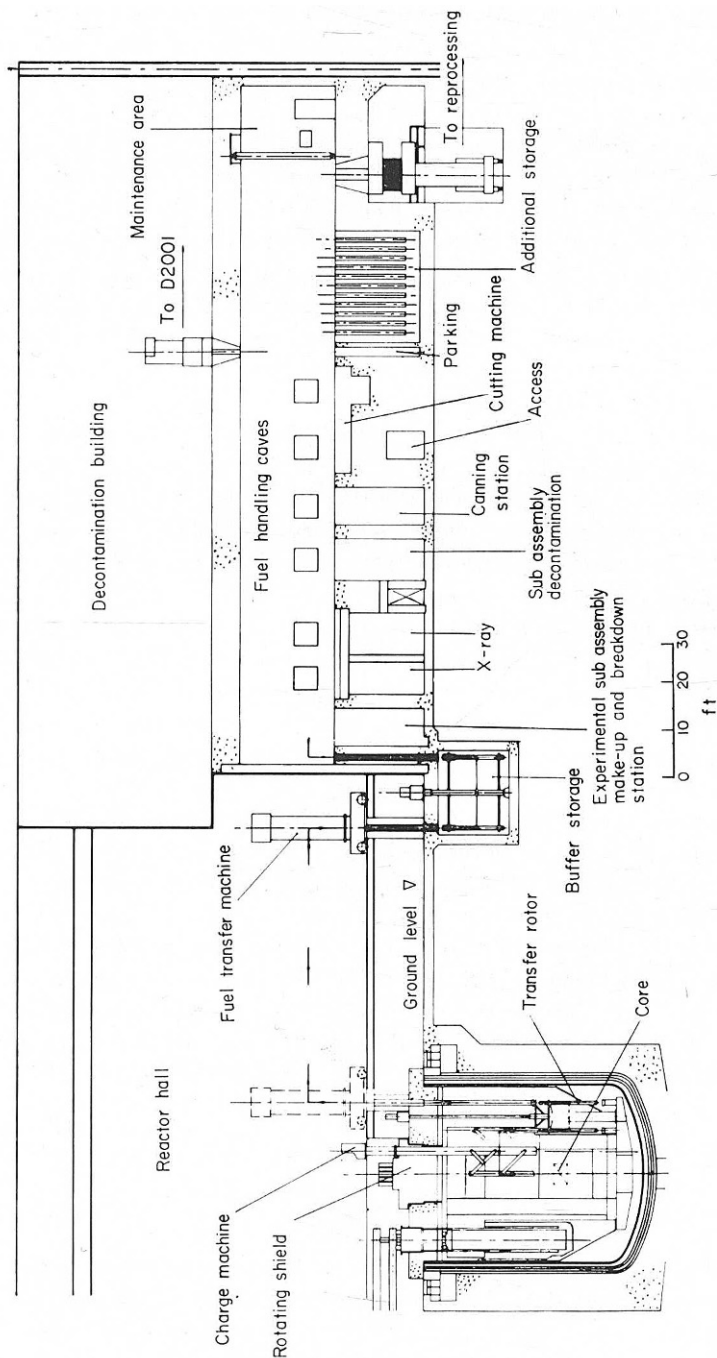


FIG. 7. Fuel handling route.

is mounted on a ball race and is provided with a mercury dip seal. The shield is cooled by a forced convection argon system and the structure is thermally insulated by a number of thin stainless steel plates placed $\frac{1}{2}$ in. (1.27 cm) apart. Shielding is provided by concrete.

The charge machine is approximately 42 ft (1280 cm) long and comprises a system of links producing parallel motion of a charge chute. The charge chute incorporates the subassembly grab unit which is moved within a tube by means of a ball nut and screw mechanism. In addition to supporting the gripper mechanism, the charge chute tube provides restraint on adjacent subassemblies during subassembly withdrawal and transmits the withdrawal load directly to the grid. Latching and delatching of the grab is controlled by cams on the charge chute tube and delatching can only take place when the grab is in the lower position.

The charge chute can be traversed radially inwards to the reactor centre line, and following rotation of the charge machine through 180° may be traversed outwards as shown in Fig. 7 through a slot in the radial neutron shield to a position above the charge-discharge port. This port is formed in the thermal insulation separating the hot and cold sodium and allows access to the fuel transfer rotor.

The charge machine will be removed from the reactor after completing the charging cycle and either stored in a leak tight flask filled with inert gas or decontaminated, stripped down and reconditioned in readiness for the next fuel cycle. To ease the problems of removal, the machine is designed in two sections, the first section being easily detachable and requiring no flask for removal.

The transfer rotor provides storage within the reactor vessel allowing subassemblies to lose decay heat to the primary sodium for about 30 days, by which time the decay heat is approximately 12 kW. It also enables fuel movements into and out of the reactor to be carried out through a single transfer port located in the roof, clear of the rotating shield. The rotor consists of a ribbed circular disk, 228 cm o.d., suspended on a shaft from the reactor roof. Twenty storage positions are provided for stainless steel buckets which are used to transfer the decayed subassemblies from the reactor vessel. Positive checks are provided prior to rotating the drum, to ensure that disconnexions have been completed.

(g) Post-irradiation and New Fuel Handling

Subassemblies, in buckets, are withdrawn from the storage rotor into a cast iron flask. The flask which accommodates a single subassembly has no forced gas cooling and operates on a rail track linking the transfer port with the buffer storage and the new fuel store, as indicated in Fig. 7. There is

no special handling route for failed subassemblies as all discharged fuel is accommodated in sodium filled buckets.

The buffer storage consists of a rotating drum operating in a NaK filled vessel. The drum provides access for fuel into an argon filled cave, in which facilities are provided for assembling and breaking down experimental fuel assemblies, radiography, metrology, and visual examination and decontamination of complete subassemblies.

New fuel subassemblies may be loaded directly from the new fuel store into a single subassembly flask and routed to the reactor transfer port. Alternatively, new fuel may be loaded into empty buckets at the buffer storage and then via a cast iron flask into the reactor storage drum.

(h) PFR Building Layout

The buildings provide conditions in which the plant can be operated safely and efficiently. The objective has been to design buildings which are economic to construct and are pleasing in appearance. In addition, since considerable mechanical site fabrication is necessary, the buildings and the construction programme have been designed to allow the maximum use of the buildings and permanent services during the construction period.

A compact layout has been adopted with all plant housed in one building complex, the complex including a central change room, a control room, and administration block. By this arrangement, piping services and cable routes, staff requirements and operating costs are a minimum and construction costs are less than for a group of separate buildings. The complete layout is shown in outline in Fig. 8.

The reactor hall is a low pressure containment building designed to withstand a maximum internal pressure of 20 in. W.G. (50.8 mb) with a reasonably low leak rate. The building is of steel frame construction, and in addition to the reactor it houses the primary cold trap. Fuel is moved to and from the reactor to the fuel handling caves by a mobile flask and leaves the containment via the flask valve.

The decontamination and fuel handling building and active workshop is a building of similar dimensions and in line with the reactor hall. Equipment can be moved directly into the decontamination building from the reactor hall and the facilities provided allow for the progressive decontamination and dismantling of reactor components, their inspection, repair, and re-erection. Steam cleaning followed by water wash is the standard method of decontamination but facilities for alcohol wash for delicate components are provided. The building also houses a post-irradiation cave facility for examination, decontamination, and storage of spent fuel together with an off-loading station for the dispatch of fuel for further examination and/or processing. A new fuel storage and inspection area is included. The reactor hall and

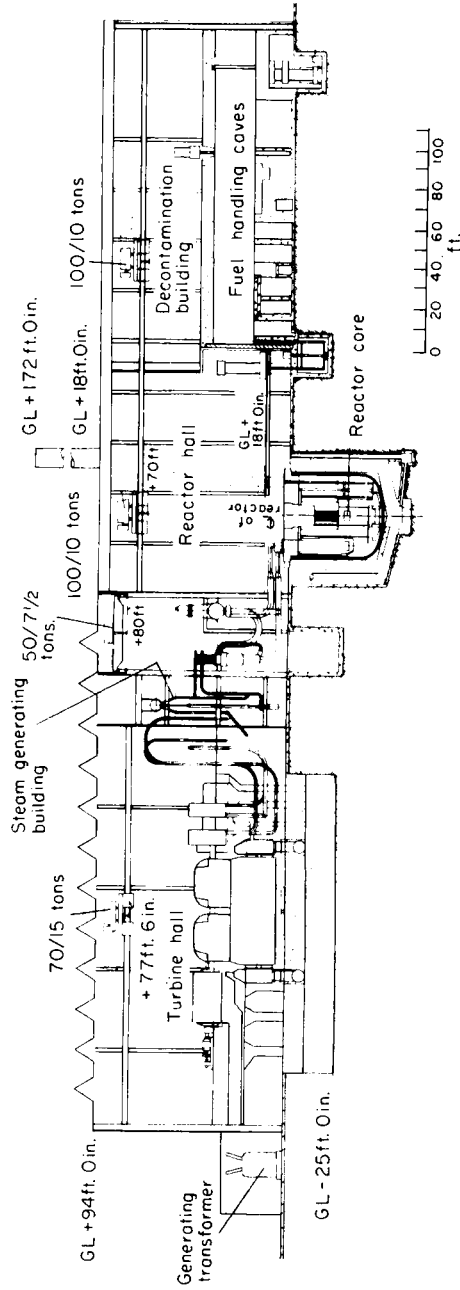


FIG. 8a. Longitudinal section.

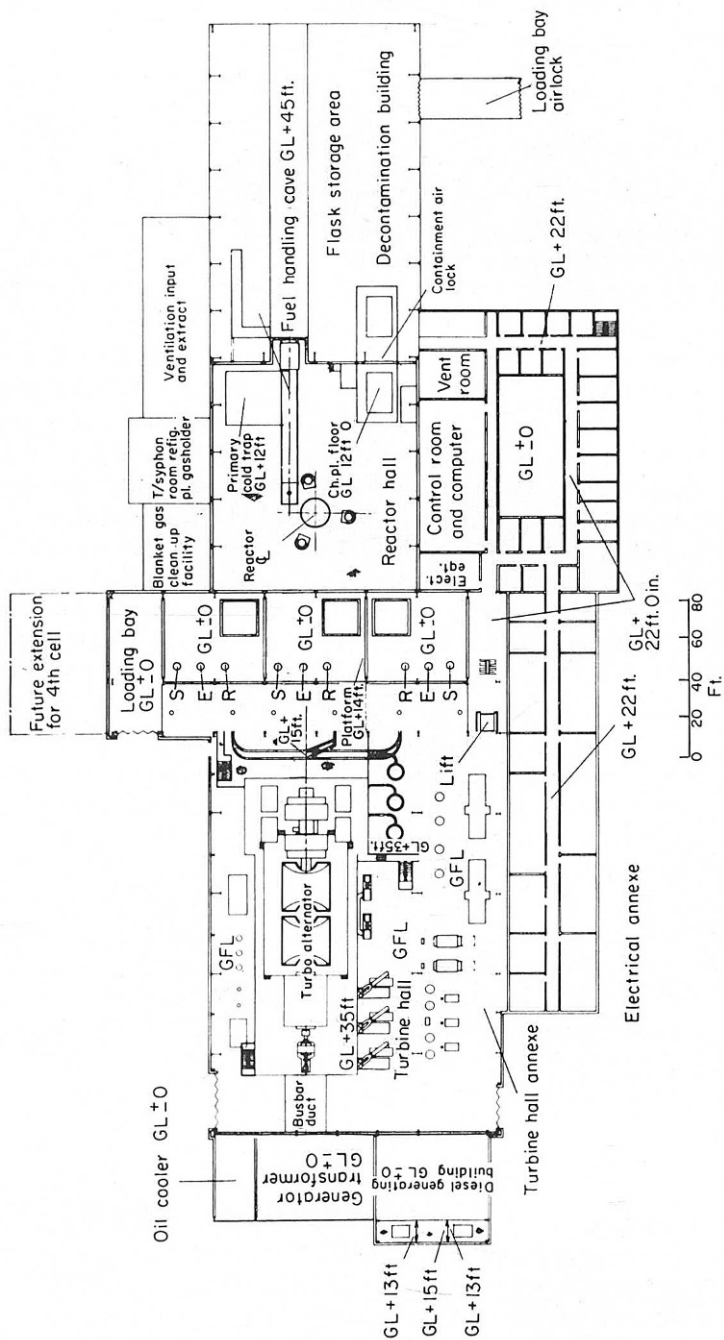


FIG. 8b. General plan.

decontamination building and crane facilities will be used for the construction of vessel, vault roof, and vessel internals.

4. Safety Considerations

(a) Subassembly Accidents

In the scarcely credible event of undetected failure of all the fuel pins in a subassembly, failure cannot propagate by melting across two adjacent subassembly wrappers. The worst reactivity effects, i.e. compaction of all fuel within a subassembly combined with sodium voiding, will not give rise to an unacceptable core transient. After shutdown decay heat from a blocked subassembly can be removed without internal boiling by coolant flowing in adjacent subassemblies. In the absence of propagation of a failure from one subassembly to another, the subassembly has been further examined in relation to a postulated accident in which coolant is separated from the fuel and later regains contact with an explosive generation of sodium vapour.

Analysis of this incident after a hypothetical initiating blockage shows that a minimum amount of fuel, equivalent to ten fuel pins, is required before sufficient energy is available to disrupt the blocked subassembly wrapper, ignoring any constraint from its neighbours. It is also clear that however much fuel is involved, thermal potential for the conversion of heat energy into destructive mechanical energy cannot be built up in less than 1.5 sec after melting of the cans. In the most pessimistic view, therefore, if a failure involving more than ten pins can be detected and the reactor shut down within 1.5 sec of melting of the cans this type of incident cannot develop. Failure to detect a fault involving less than ten fuel pins can only result in an incident which will be confined within one subassembly.

Development of fast acting detection systems is in hand⁽²⁾ but at the moment only reactivity changes can be relied upon to trip the reactor within $1\frac{1}{2}$ sec and these can only be ambiguously related to pin failure. Core thermocouples are provided and although they will trip the reactor within a few seconds, they will not be sensitive to small faults. Delayed neutron monitors are proposed and are expected to indicate quite small failures within 10–15 sec. It is considered unlikely that a fault will propagate to ten complete pins in this timescale but it has not been possible to establish this by analysis.

(b) Failure Criteria

As data become available on the properties of irradiated canning material, failure criteria are kept under review. The present position is that failure of fully irradiated pins may take place due to creep rupture if the can temperature exceeds 750°C for a long period and due to loss of ultimate strength if the can temperature should exceed 800–850°C.

(c) Reactivity

Two sets of absorber rods are provided—control rods to control reactivity changes related to power, temperatures and burnup, and shut-off rods to provide adequate shutdown during fuel changing and for rapid termination of fault excursions. The rate of rod withdrawal is controlled to give an estimated maximum rate of reactivity addition of \$4 per minute assuming simultaneous withdrawal of both control and shut-off rods, an error which is safeguarded against by interlocks but which transient analysis shows would not lead to conditions in excess of the failure levels. Fuel changing is carried out with the reactor shut down and all absorber inserted into the core and disconnected from its actuators. Reactivity changes during fuel handling are indicated by sub-critical reactivity monitors.

(d) Coolant Flow

Electrical supplies to the three primary pumps are segregated as far as possible. The worst flow failure conditions envisaged arise from seizure of one pump (assumed instantaneous) or run-down of all pumps. There is sufficient inertia in the rotating pumps to allow tripping of the reactor by outlet temperature, flow change or flux-flow ratio trip signals without exceeding the failure criteria. The operating point when running at reduced power on two pumps will be chosen such that the transient resulting from these failure conditions can be accepted. Automatic non-return valves are not proposed. For decay heat removal, circulation of primary sodium is provided by pony motors driving the main pumps and fed from a guaranteed non-interruptable supply. Heat rejection from the primary coolant is not required for several hours due to its thermal capacity. Heat rejection can take place, however, via natural circulation removal heat exchangers.

(e) Fission Product Containment

Leakage of activity from the primary containment following a core melt-down is minimized by controlling the internal pressure to a low value. This is done by the normal blanket gas cleanup system which can accommodate the incondensable gases displaced from the primary circuit and can also condense any vapour which may be produced. Any leakage of active products through the vault roof takes place into the main reactor building which is a low-pressure building built to a reasonably high standard of leak tightness and provided with a chemical cleanup plant. It is designed to contain the pressure rise due to combustion of a transfer bucket full of sodium resulting from a fuel handling accident, and would give an additional decontamination factor in the event of a core incident.

5. Experimental Facilities

The experimental facilities provided include for:

(i) *Fuel development*

Core positions have been allocated for special subassemblies with permanently connected instrumentation. A hexagonal subassembly, 8.25 in. across flats, can be tested by replacing a number of normal core subassemblies. Vented and subdivided fuel subassemblies can also be tested. Complete post-irradiation examination facilities are provided.

(ii) *Coolant purity instrumentation*

A number of sodium purity instruments in excess of normal operational requirements are provided. These include plugging meters, rhometers, corrosion monitors and sodium sampling facilities.

(iii) *Alternative steam generator concepts*

The steam generator building has been designed so that an additional bay can be added to test an alternative arrangement of boiler.

Acknowledgement

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APPENDIX

LIST OF MAIN PARAMETERS

Thermal output	559 MW core	41 MW breeder
Gross electrical output from generator	265 MW approx.	
Net electrical output from station	248 MW approx.	
Coolant temperature at core inlet	400–430°C	752–806°F
Coolant temperature at core outlet	560–600°C	1040–1110°F
Secondary sodium temperature at steam generator inlet	530°C	986°F
Secondary sodium temperature at steam generator outlet	370°C	698°F
Primary flow through core	23.2×10^6 lb/h	1.76×10^5 kg/min
Primary pump head	117 lb/in ²	8.23 kg/cm ²
Secondary coolant flow	23.2×10^6 lb/h	1.76×10^5 kg/min
Secondary pump head	45 lb/in ²	3.15 kg/cm ²
No. of fuel pins per subassembly	325	
Outside diameter of fuel can	0.23 in.	0.585 cm
Fuel can thickness	0.015 in.	0.038 cm
No. of subassemblies in inner zone	30	
No. of subassemblies in outer zone	48	
No. of radial breeder subassemblies	42	
Length of fuel in pin	36 in.	91.5 cm
Length of axial breeder	9 in.	22.9 cm
Overall length of pin	103 in.	262 cm
Weight of mixed oxide fuel in core	4.0 tonnes	
Weight of PuO ₂ in core	0.902 tonne	
Diameter of primary vessel	40 ft.	1221 cm
Weight of sodium in primary vessel	900 tonnes	
Steam temperature at T.S.V.	513–566°C	955–1050°F
Steam pressure at T.S.V.	2315 lb/in ² a	153 kg/cm ²
Reheat pressure inlet	495 lb/in ² a	34.5 kg/cm ²
Feed temperature	288°C	550°F